

Charalambos Mitropoulos

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Research Interests

My research interests lie in the area of smart contracts, program analysis, and software testing.

Education

PhD in Computer Science – Technical University of Crete Sept 2020 – July 2026

Thesis: Strengthening the Solidity Ecosystem Through Empirical Analysis and Feedback-Driven Methods

Supervisor: Prof. S. Ioannidis

Indicative courses and grades:

- Advanced Computer Architecture: 10/10
- Computer and Systems Security: 7/10

MSc in Information Systems – Athens University of Economics and Business Sept 2017 – May 2019

GPA: 8/10

Thesis: Employing Different Program Analysis Methods to Study Bug Evolution

Supervisor: Prof. D. Spinellis

Indicative courses and grades:

- Software Reliability and Quality Assurance: 8/10
- Computer Security: 8/10
- Cryptography: 9/10

BSc in Informatics and Telecommunications – National and Kapodistrian University of Athens Sept 2011 – Sept 2017

GPA: 7/10

Thesis: Local Search for Optimization Problems

Supervisor: Prof. V. Zissimopoulos

Indicative courses and grades:

- Operating Systems: 8/10
- Algorithms and Complexity: 8/10
- Systems Programming: 7/10

High School Diploma Sept 2011

GPA: 18.5/20

Awards and Scholarships

Alpha Bank's Scholarship for First-Year Students 2012-2013

A scholarship provided by the bank to students who have GPA above 7.5/10

SIGSOFT CAPS August 2019

Student Travel Support and Childcare Assistance at Conferences. The grant was given by the organizing committee of ESEC/FSE '19.

Experience

Research Associate January 2019 – Today

Business Analytics Lab ([Balab](#) ) , Athens University of Economics and Business

Research on software security and testing

Web Development, IT Support June 2016 – March 2026

Development and support of the [Dikotylon](#)  website.

Technologies: HTML, JavaScript

Researcher at Foundation of Research and Technology – Hellas

August 2020 – December

Participated in European funded programmes:

2021

- ASCAPE
- SPIDER 5G CYBER RANGE

Researcher at Technical University of Crete

December 2021 –

Participated in European funded programmes:

December 2025

- CyberSecPro
- Sentinel
- SecOpera
- Cybersecdome

Partner at DIENEKES

January 2026 – Today

Participated in European funded programmes:

- SYNAPSE (EU Award Number: 101120853)

Publications

- **Charalambos Mitropoulos**, Maria Kechagia, Chrysostomos Maschas, Sotiris Ioannidis, Federica Sarro, and Dimitris Mitropoulos. [Broken agreement: The evolution of Solidity error handling](#) [↗](#). In *Proceedings of the 18th ACM/IEEE International Symposium on Empirical Software Engineering and Measurement (ESEM '24)*. Pages 257–268. October 2024.
- **Charalambos Mitropoulos**, Thodoris Sotiropoulos, Sotiris Ioannidis, and Dimitris Mitropoulos. [Syntax-Aware Mutation for Testing the Solidity Compiler](#) [↗](#). In *Proceedings of the 28th European Symposium on Research in Computer Security (ESORICS '23)*. Pages 327–347. Springer. September 2023.
- Stefanos Chaliasos, Thodoris Sotiropoulos, Georgios-Petros Drosos, **Charalambos Mitropoulos**, Dimitris Mitropoulos, and Diomidis Spinellis. [Well-typed programs can go wrong: a study of typing-related bugs in JVM compilers](#) [↗](#). In *Proceedings of the ACM on Programming Languages, Volume 5, Issue OOPSLA (part of SPLASH 2021)*. Article No. 123. Pages 1–30. October 2021.
- **Charalambos Mitropoulos**. [Employing Different Program Analysis Methods to Study Bug Evolution](#) [↗](#). In *Proceedings of the 27th ACM Joint Meeting on European Software Engineering Conference and Symposium on the Foundations of Software Engineering (ESEC/FSE '19): Student Research Competition (SRC) Track*. Pages 1202–1204. August 2019.

PC Member

- 2025: ISSTA (Artifact Evaluation Committee)
- 2026: TSE Journal (Reviewer)

Teaching Assistant

Computer Security. Technical University of Crete. BSc course, 5th semester. 2022–Today.

Talks

- **Seminar Presentation @ AUEB's BALAB (2023):** Syntax-Aware Mutation for Testing the Solidity Compiler
- **PL-Seminar @ National Technical University of Athens, Software Engineering Laboratory (2023):** Syntax-Aware Mutation for Testing the Solidity Compiler
- **Seminar Presentation @ AUEB's BALAB (2024):** Broken Agreement: The Evolution of Solidity Error Handling

Personal Projects

Testing the Solidity Compiler:

Fuzzing the Solidity compiler (Solidity is an object-oriented language for writing smart contracts) with AFL (American Fuzzy Lop). The upstream developers provided instant bug-fix releases after approving my patches. Some of these are:

- ICE on missing modifier identifier (#7550) [↗](#)
- ICE on array index with ABIEncoderV2 (#7549) [↗](#)
- ICE when assigning to calldata structs and static arrays using inline assembly; FixedPointType not-implemented error (#11680) [↗](#)
- ICE in CompilerUtils::convertType when converting a calldata slice of bytes or string (#11677) [↗](#)

Testing the Zero-Knowledge Compilers:

Testing Zero-Knowledge compilers (Circom, Cairo, Noir, Leo) for security issues. The upstream developers provided instant bug-fix releases after approving my patches. Some of these are:

- Out-of-bounds array access in witness generation returns arbitrary values without error [↗](#)
- Boolean value true silently coerced to field element in CLI arguments [↗](#)
- Hexadecimal inputs (0xff) silently accepted without documentation [↗](#)
- Compiler stack overflow crash on wide expressions (~1500 terms) [↗](#)

Technical Skills

Programming Languages: C, C++, Java, SQL, Python, PHP, Prolog, Scala, HTML, JavaScript, Bash scripting

Tools and Editors: Vim, IntelliJ, Eclipse, NetBeans, Hadoop, MySQL, MySQL Workbench

Documentation and Preparation Systems: LaTeX, Word, Excel, PowerPoint

Foreign Languages

English: Certificate of Proficiency in English, University of Michigan.

German: Goethe-Zertifikat A